



HABSHAN & DIVERT EXCESS MP FUEL GAS PROJECT

AGREEMENT NO. 4700024475

RAD PROJECT No: 5900982

AREJLERS



COMMENTS RESOLUTION SHEET

ADNOC GAS

Document No:	NM-30201-50200-H0-113-13-15-25-001			Rev:	D
Document Title:	INSTRUMENT I/O LIST			Doc. Class:	2
ACONEX Workflow Transmittal No.:	ADNOCGP-WTRAN-001392			Date	12-12-2025
Comment Sheet No:	2, 3, 4			Rev:	0
Document Status Code:	3	Code 1 – Document Rejected Code 2 – Comments as Noted	Code 3 – No Comments/ No Further Comments Code 4 – Information Only		
S.No.	COMPANY Comments (Identify Section/Page/Paragraph) Rev. C/ Date: 12-05-2025	CONTRACTOR / VENDOR Reply Rev. C/ Date: 14-10-2025		COMPANY Decision	
1.	Page 2 of 4 signals shall be configured/mapped on the existing serial interface between ESD and DCS in line with existing plant philosophy. 4. Alarm Priorities and Rationalization shall be done for the Project SOW in line with existing Plant Philosophy. DOCUMENT	COMPANY Comment is noted and document has been updated accordingly.			
2.	Page 2 of 4 SDOC-5225-01-1510-0010-0036 I/O system Map-Modus interchange Include Alarm & Trip Schedule also in Reference Document for completeness.	COMPANY Comment is noted and document has been updated accordingly.			
3.	Page 2 of 4  ?? Replace with "4" . (DCS Open & Close commands I-O shall be considered as DO in DCS and DI in ESD System.)	COMPANY Comment is noted and document has been updated accordingly.			



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4.	Page 3 of 4 	COMPANY Comment is noted and document has been updated accordingly.	
5.	Page 4 of 4 	COMPANY Comment is noted and document has been updated accordingly.	
COMMENT ON REV.C			



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1.	Page 2 of 4 Temp. : Pressure I/O SUMMARY LIST: <table><thead><tr><th></th><th>DCS Signals</th><th>ESD Signals</th></tr><tr><th>I/O Type</th><th>Qty.</th><th>Qty.</th></tr></thead><tbody><tr><td>AI (IS)</td><td>5</td><td>4</td></tr><tr><td>AO (IS)</td><td>1</td><td>1</td></tr><tr><td>DI (IS)</td><td>-</td><td>4</td></tr><tr><td>DI-NIS</td><td>-</td><td>4</td></tr><tr><td>DO</td><td>2</td><td>5</td></tr></tbody></table> replace with "3" as Partial stroke testing PT is not applicable for main ESDV as per latest requirement. replace with "." as Partial stroke testing is not applicable for Main ESDV as per latest requirement. Replace with "8" as per latest Company End Users requirement. ?? Replace with "2" . (This DCS Close command I-O shall be considered as DO in DCS and DI in ESD System. replace with "2" as Partial stroke testing is not applicable for main ESDV as per latest requirement.		DCS Signals	ESD Signals	I/O Type	Qty.	Qty.	AI (IS)	5	4	AO (IS)	1	1	DI (IS)	-	4	DI-NIS	-	4	DO	2	5	The I/O list table has been in accordance with the company instructions.	
	DCS Signals	ESD Signals																						
I/O Type	Qty.	Qty.																						
AI (IS)	5	4																						
AO (IS)	1	1																						
DI (IS)	-	4																						
DI-NIS	-	4																						
DO	2	5																						
2.	Page 2 of 4 <table><tr><td>D-SC</td><td>2</td><td>-</td></tr></table> ?? What is the basis? Check & clarify. Correct as applicable.	D-SC	2	-	Since there is no PST required for the shutdown valve, the I/O "D-SC" is removed from the I/O Summary list table in the updated Rev-D document.																			
D-SC	2	-																						
3.	Page 3 of 4 Alarm Priority shall be finalized and indicated by Detail Design Consultant in the I-O List for completeness (Previous Revision comment not addressed)	Noted, the Alarm and Trip Settings are updated Accordingly Doc.No : DOD-30201-50200-H0-113-13-00-00-006																						
4.	Page 3 of 4 DCS & ESD I-O List shall be provided in separate sheets with segregation for clarity and traceability. Further, these are to be executed by different vendors under different projects and so it is a mandatory requirement to have segregation of DCS & ESD I-O lists (Previous Revision comment not addressed)	Noted, the DCS & ESD tags are provided separately in the updated I/O list document, Rev-D																						



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5.	Page 3, 4 of 4	<table><thead><tr><th>I/O Type</th><th>IS / NIS</th><th>Wire Type</th><th>P&ID No</th><th>Sys. R</th></tr><tr><th>System</th><th>Signal Type</th><th>Wet / Dry</th><th>Device Location</th><th>Min.</th></tr><tr><th>HMI Tag</th><th>Voltage LVL</th><th>NO / NC</th><th>MOS</th><th>OOS</th></tr></thead><tbody><tr><td>D-AI</td><td>IS</td><td>2W(HART)</td><td rowspan="3">Incomplete. Check & include all applicable HMI tags for all I-Os listed.</td><td></td></tr><tr><td>DCS</td><td>4-20 mA</td><td></td><td></td></tr><tr><td>113-FI -1851</td><td>24V DC</td><td></td><td></td></tr><tr><td>D-AI</td><td>IS</td><td>2W(HART)</td><td>113-13-00-00-011</td><td>0</td></tr><tr><td>DCS</td><td>4-20 mA</td><td></td><td>FIELD</td><td>ba</td></tr><tr><td></td><td>24V DC</td><td></td><td></td><td></td></tr><tr><td>D-AO</td><td>IS</td><td>2W(HART)</td><td>113-13-00-00-011</td><td></td></tr><tr><td>DCS</td><td>4-20 mA</td><td></td><td>FIELD</td><td></td></tr><tr><td></td><td>24V DC</td><td></td><td></td><td></td></tr><tr><td>D-AI</td><td>IS</td><td>2W(HART)</td><td>113-13-00-00-011</td><td>0</td></tr><tr><td>DCS</td><td>4-20 mA</td><td></td><td>FIELD</td><td>%</td></tr><tr><td>113-PZ1 -3191</td><td>24V DC</td><td></td><td></td><td></td></tr><tr><td>D-AI</td><td>IS</td><td>2W(HART)</td><td>113-13-00-00-011</td><td>0</td></tr><tr><td>DCS</td><td>4-20 mA</td><td></td><td>FIELD</td><td>ba</td></tr><tr><td></td><td>24V DC</td><td></td><td></td><td></td></tr><tr><td>E-AI</td><td>IS</td><td>2W(HART)</td><td>113-13-00-00-011</td><td>0</td></tr><tr><td>ESD</td><td>4-20 mA</td><td></td><td>FIELD</td><td>ba</td></tr><tr><td></td><td>24V DC</td><td></td><td>Yes</td><td></td></tr><tr><td>E-AI</td><td>IS</td><td>2W(HART)</td><td>113-13-00-00-011</td><td>0</td></tr><tr><td>ESD</td><td>4-20 mA</td><td></td><td>FIELD</td><td>ba</td></tr><tr><td></td><td>24V DC</td><td></td><td>Yes</td><td></td></tr><tr><td>E-AI</td><td>IS</td><td>2W(HART)</td><td>113-13-00-00-011</td><td>0</td></tr><tr><td>ESD</td><td>4-20 mA</td><td></td><td>FIELD</td><td>ba</td></tr><tr><td></td><td>24V DC</td><td></td><td>Yes</td><td></td></tr><tr><td>D-AI</td><td>IS</td><td>2W(HART)</td><td>113-13-00-00-011</td><td>-4</td></tr><tr><td>DCS</td><td>4-20 mA</td><td></td><td>FIELD</td><td></td></tr></tbody></table>	I/O Type	IS / NIS	Wire Type	P&ID No	Sys. R	System	Signal Type	Wet / Dry	Device Location	Min.	HMI Tag	Voltage LVL	NO / NC	MOS	OOS	D-AI	IS	2W(HART)	Incomplete. Check & include all applicable HMI tags for all I-Os listed.		DCS	4-20 mA			113-FI -1851	24V DC			D-AI	IS	2W(HART)	113-13-00-00-011	0	DCS	4-20 mA		FIELD	ba		24V DC				D-AO	IS	2W(HART)	113-13-00-00-011		DCS	4-20 mA		FIELD			24V DC				D-AI	IS	2W(HART)	113-13-00-00-011	0	DCS	4-20 mA		FIELD	%	113-PZ1 -3191	24V DC				D-AI	IS	2W(HART)	113-13-00-00-011	0	DCS	4-20 mA		FIELD	ba		24V DC				E-AI	IS	2W(HART)	113-13-00-00-011	0	ESD	4-20 mA		FIELD	ba		24V DC		Yes		E-AI	IS	2W(HART)	113-13-00-00-011	0	ESD	4-20 mA		FIELD	ba		24V DC		Yes		E-AI	IS	2W(HART)	113-13-00-00-011	0	ESD	4-20 mA		FIELD	ba		24V DC		Yes		D-AI	IS	2W(HART)	113-13-00-00-011	-4	DCS	4-20 mA		FIELD		Noted, the I/O list document has been revised as per the company comments and updated.	
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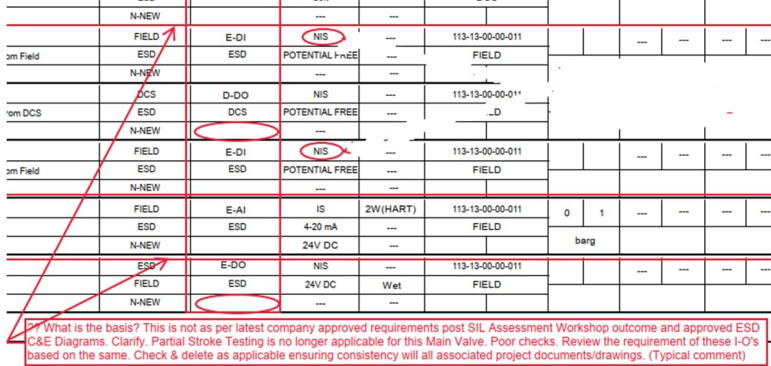
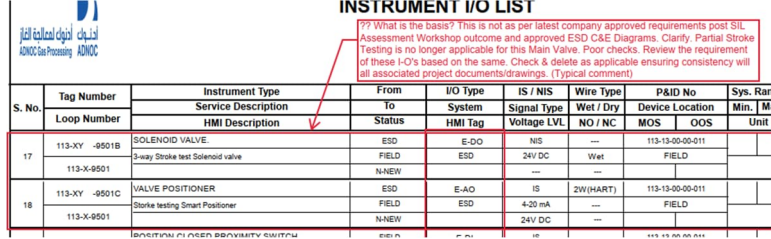
COMMENTS RESOLUTION SHEET

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6.	Page 3, 4 of 4 Partial Stroke Test Enable for 113-XV-9501 501B STROKE TEST RESET Partial Stroke Test Reset for 113-XV-9501 501A HAND SWITCH CLOSE 113-XV-9501 Valve Close HS command From Field 501B HAND SWITCH CLOSE 113-XV-9501 Valve Close HS command From DCS 113-XV-9501 Valve Closed Position limit switch POSITION OPEN PROXIMITY 113-XV-9501 Valve Open Position limit switch HAND SWITCH CLOSE 113-XV-9502 Valve Close HS command From Field DCS HAND SWITCH 113-XV-9502 Valve Close HS from DCS	Noted, the I/O list document has been revised as per the company comments. Since we follow up company's existing SPI Database Instrument type, we using the same.	
7.	Page 3 of 4 113-XV-9501 SOLENOID VALVE 3-way ESD trip Solenoid valve 113-XV-9501 Delete Tag no. suffix as Partial stroke testing is no longer applicable for this main valve Solenoid Valve	Noted, the tag number is corrected in the updated I/O list document, Rev-D.	

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11.	Page 3 of 4 	Noted, the tags have been removed from the I/O list loop, and the document has been revised as per the company comments.	
12.	Page 4 of 4 	Noted, the tags have been removed from the I/O list and the document has been revised as per the company comments.	



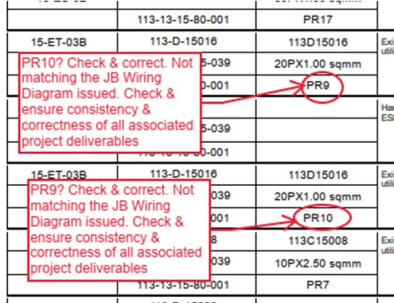
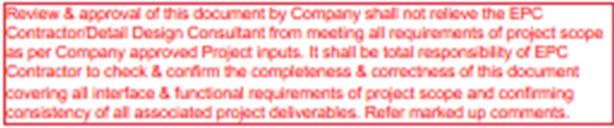
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13.	Page 4 of 4 	Noted, the loop diagram and Terminal details have been revised as per the company comments.	
COMMENT ON REV.B			
1.	(Page 1 of 5) 	COMPANY Comment is noted and agreed.	ok



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2.	(Page 2 of 5) INT NO. DESCRIPTION <table><tr><td>1</td><td>COVER SHEET</td></tr><tr><td>2</td><td>INDEX, GENERAL NOTES, ABBREVIATION & REFERENCE DOCUMENT</td></tr><tr><td>3</td><td>I/O LIST</td></tr></table> GENERAL NOTES 1) All ESD signals shall be terminated in existing junction box as per JB termination (DW-01-15-30-00) for IES-15, 15.EE-02 ESD system interfacing. 2) All DCE signals shall be terminated in IES-15 NEW DCE (Marshalling Cabinet). Existing DCE (VABE) is getting replaced with new Yokogawa Centium VP DCE as part of on-going project (P5321) HBCB-1. There is no requirement of engaging the existing DCE (VABE) (DCE) into Yokogawa modifications (addition of I/O modules, configuration of graphics, logic pertaining to engineering) shall be carried out in other on-going project (HBCB-1, P5321), modification in DCE shall be in accordance with project requirements in new DCE systems. 3) DCE Controller shall consider cable termination from feed up to the marshalling cabinets in IES. Loop resistance reference to functional logic/signals (P&ID/AT&T) should refer to I/O. 4) Serial interface signals shall be configured/trapped on the existing serial interface between ESD and DCE in accordance with existing plant philosophy. REFERENCE DOCUMENT 800-30201-50200-H0-113-13-15-00-001 - INSTRUMENTATION DESIGN BASIS 30201-50200-H0-00-0001-A1 - P&ID AREA 90 FUEL GAS & FLARE SYSTEM CONSTRUCTION 30201-50200-H0-113-13-00-0001-A1 - HABSHAN EXCESS MP FUEL GAS LINE CONSTRUCTION 30201-50200-H0-113-00-000-002-A1 - MP GAS MANIFOLD UNIT 113 (NORTH AREA) CONSTRUCTION 30201-50200-H0-13-00-210-A1 - P&ID FLARE HEADER EARLY THAMMAHA 'W' (HABSHAN) CONSTRUCTION 30201-50200-H0-13-00-210-A1 - P&ID FLARE HEADER EARLY THAMMAHA 'W' (HABSHAN) DEMOLITION 30201-50200-H0-00-00010-A1 - P&ID AREA 90 FUEL GAS & FLARE SYSTEM DEMOLITION Which Contractor, Not clear. Check & update. Check & correct, Delete, Poor Quality checks COMPANY Comment is noted and document has been updated accordingly. Noted	1	COVER SHEET	2	INDEX, GENERAL NOTES, ABBREVIATION & REFERENCE DOCUMENT	3	I/O LIST
1	COVER SHEET						
2	INDEX, GENERAL NOTES, ABBREVIATION & REFERENCE DOCUMENT						
3	I/O LIST						
3.	(Page 2 of 5) ABBREVIATION IES : Instrument Equipment Shelter DCE / ESD : Distributed Control System / Emergency Shutdown System AO : Analog Output AI : Analog Input DI : Digital Input DIP : Digital Input: Pusher Type DO : Digital Output I/O : Input/Output IS : Intrinsically Safe NI : Non Intrinsically Safe L : Low Alarm LL : Low Low Alarm H : High Alarm Update abbreviations in line with this I/O List specifically. Refer marked up comments. COMPANY Comment is noted and document has been updated accordingly. Noted						



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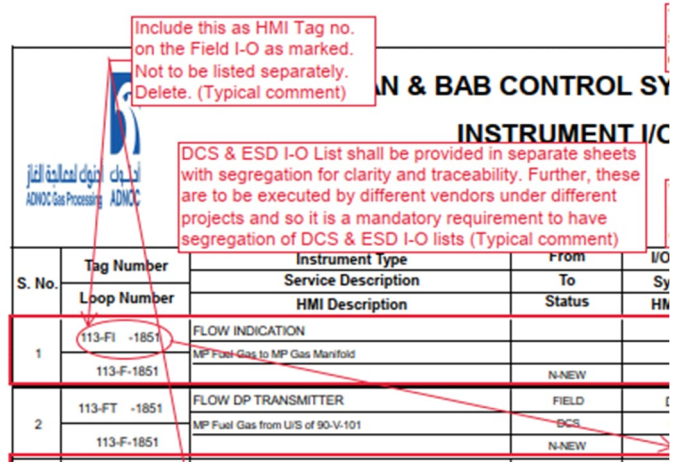
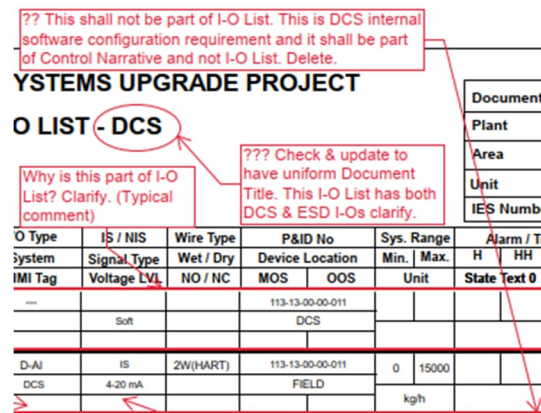
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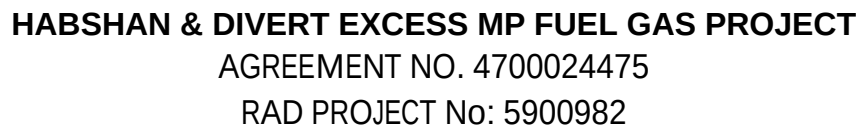
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4.	(Page 2 of 5) IO SUMMARY LIST: <table><thead><tr><th>IO Type</th><th>Qty.</th><th>SCS Signals</th><th>SSD Signals</th></tr></thead><tbody><tr><td>AI (SE)</td><td>5</td><td></td><td>4</td></tr><tr><td>AO (SE)</td><td>1</td><td></td><td>1</td></tr><tr><td>DI</td><td>-</td><td></td><td>6</td></tr><tr><td>DI-P</td><td>-</td><td></td><td>4</td></tr><tr><td>DI</td><td>2</td><td></td><td>3</td></tr><tr><td>Serial DI/DO (N/S)</td><td>10</td><td></td><td>10</td></tr></tbody></table> DI- (NIS) This I/O Summary is just noted for information only. It shall be total responsibility of the EPC Contractor / Detail Design Consultant to check & confirm the completeness & correctness of this I/O Summary as per Company approved Project inputs. (Typical comments)	IO Type	Qty.	SCS Signals	SSD Signals	AI (SE)	5		4	AO (SE)	1		1	DI	-		6	DI-P	-		4	DI	2		3	Serial DI/DO (N/S)	10		10	COMPANY Comment is noted and document has been updated accordingly. IO count rechecked and updated as per latest IO list.	Refer marked up Comments
IO Type	Qty.	SCS Signals	SSD Signals																												
AI (SE)	5		4																												
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DI	2		3																												
Serial DI/DO (N/S)	10		10																												
5.	(Page 2 of 5) HOLD: 1) Ranges mentioned in IO list are Preliminary. However same shall be verified during Vendor detailed engineering and if required same shall be updated in upcoming revisions. All Holds shall be removed for final approval	COMPANY Comment is noted and agreed.	Noted																												

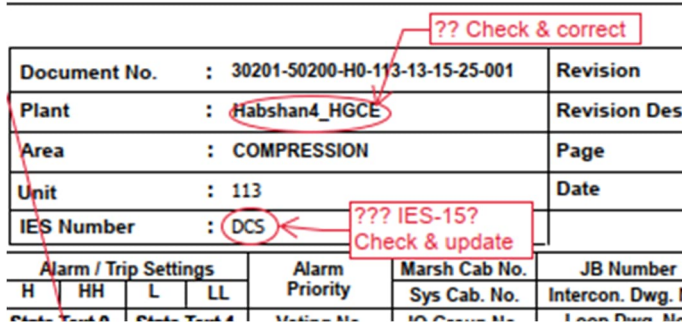
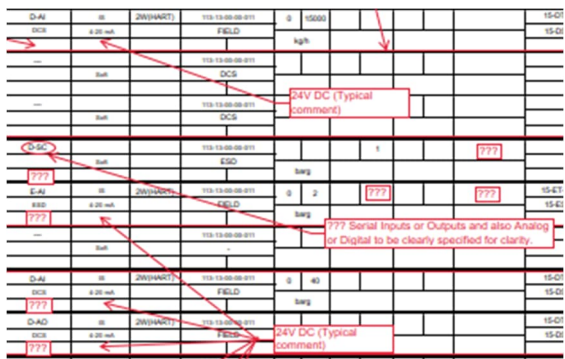
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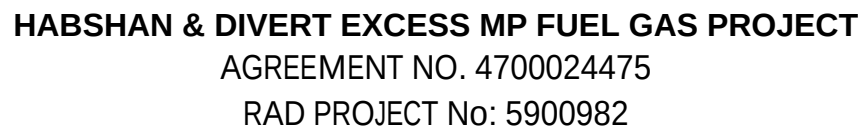
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6.	(Page 3 of 5) Include this as HMI Tag no. on the Field I-O as marked. Not to be listed separately. Delete. (Typical comment) DCS & ESD I-O List shall be provided in separate sheets with segregation for clarity and traceability. Further, these are to be executed by different vendors under different projects and so it is a mandatory requirement to have segregation of DCS & ESD I-O lists (Typical comment)  HABSHAN & DIVERT EXCESS MP FUEL GAS PROJECT INSTRUMENT I/O LIST <table border="1"> <thead> <tr> <th>S. No.</th> <th>Tag Number</th> <th>Instrument Type</th> <th>From</th> <th>To</th> <th>I/O</th> </tr> <tr> <th></th> <th>Loop Number</th> <th>Service Description</th> <th></th> <th></th> <th></th> </tr> <tr> <th></th> <th></th> <th>HMI Description</th> <th>Status</th> <th></th> <th>Hi</th> </tr> </thead> <tbody> <tr> <td>1</td> <td>113-FI -1851</td> <td>FLOW INDICATION</td> <td></td> <td></td> <td></td> </tr> <tr> <td></td> <td>113-F-1851</td> <td>MP Fuel Gas to MP Gas Manifold</td> <td></td> <td></td> <td></td> </tr> <tr> <td></td> <td></td> <td></td> <td>N-NEW</td> <td></td> <td></td> </tr> <tr> <td>2</td> <td>113-FI -1851</td> <td>FLOW DP TRANSMITTER</td> <td></td> <td></td> <td></td> </tr> <tr> <td></td> <td></td> <td>MP Fuel Gas from U/S of 90-V-101</td> <td></td> <td></td> <td></td> </tr> <tr> <td></td> <td>113-F-1851</td> <td></td> <td></td> <td>FIELD</td> <td></td> </tr> <tr> <td></td> <td></td> <td></td> <td>DCS</td> <td></td> <td></td> </tr> <tr> <td></td> <td></td> <td></td> <td>N-NEW</td> <td></td> <td></td> </tr> </tbody> </table>	S. No.	Tag Number	Instrument Type	From	To	I/O		Loop Number	Service Description						HMI Description	Status		Hi	1	113-FI -1851	FLOW INDICATION					113-F-1851	MP Fuel Gas to MP Gas Manifold							N-NEW			2	113-FI -1851	FLOW DP TRANSMITTER						MP Fuel Gas from U/S of 90-V-101					113-F-1851			FIELD					DCS						N-NEW			COMPANY Comment is noted and document has been updated accordingly.	Refer marked up Comments
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			N-NEW																																																																		
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		MP Fuel Gas from U/S of 90-V-101																																																																			
	113-F-1851			FIELD																																																																	
			DCS																																																																		
			N-NEW																																																																		
7.	(Page 3 of 5) ?? This shall not be part of I-O List. This is DCS internal software configuration requirement and it shall be part of Control Narrative and not I-O List. Delete. Why is this part of I-O List? Clarify. (Typical comment) ??? Check & update to have uniform Document Title. This I-O List has both DCS & ESD I-Os clarify.  SYSTEMS UPGRADE PROJECT I/O LIST - DCS <table border="1"> <thead> <tr> <th>O Type</th> <th>IS / NIS</th> <th>Wire Type</th> <th>P&ID No</th> <th>Sys. Range</th> <th>Alarm / Tr</th> </tr> <tr> <th>System</th> <th>Signal Type</th> <th>Wet / Dry</th> <th>Device Location</th> <th>Min. Max.</th> <th>H HH</th> </tr> <tr> <th>IMI Tag</th> <th>Voltage LVL</th> <th>NO / NC</th> <th>MOS</th> <th>OOS</th> <th>Unit</th> </tr> </thead> <tbody> <tr> <td></td> <td></td> <td></td> <td>113-13-00-00-011</td> <td></td> <td></td> </tr> <tr> <td></td> <td>Soft</td> <td></td> <td>DCS</td> <td></td> <td></td> </tr> <tr> <td>D-AI</td> <td>IS</td> <td>2W(HART)</td> <td>113-13-00-00-011</td> <td>0 15000</td> <td></td> </tr> <tr> <td>DCS</td> <td>4-20 mA</td> <td></td> <td>FIELD</td> <td></td> <td></td> </tr> <tr> <td></td> <td></td> <td></td> <td></td> <td>kg/h</td> <td></td> </tr> </tbody> </table>	O Type	IS / NIS	Wire Type	P&ID No	Sys. Range	Alarm / Tr	System	Signal Type	Wet / Dry	Device Location	Min. Max.	H HH	IMI Tag	Voltage LVL	NO / NC	MOS	OOS	Unit				113-13-00-00-011				Soft		DCS			D-AI	IS	2W(HART)	113-13-00-00-011	0 15000		DCS	4-20 mA		FIELD							kg/h		COMPANY Comment is noted and document has been updated accordingly.	Noted																		
O Type	IS / NIS	Wire Type	P&ID No	Sys. Range	Alarm / Tr																																																																
System	Signal Type	Wet / Dry	Device Location	Min. Max.	H HH																																																																
IMI Tag	Voltage LVL	NO / NC	MOS	OOS	Unit																																																																
			113-13-00-00-011																																																																		
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D-AI	IS	2W(HART)	113-13-00-00-011	0 15000																																																																	
DCS	4-20 mA		FIELD																																																																		
				kg/h																																																																	



ADNOC GAS

(Page 3 of 5) 	COMPANY Comment is noted and document has been updated accordingly. However COMPANY to note that we are working on COMPANY Live SPI database where the existing PSR is following for the project scope. Plant name will be as Habshan4_HGCE only, SPI Database name can't modified for same. IES-15 can not be updated in SPI databse due to SPI limitation that is why we kept this as blank. Refer General notes 1 & 2 for IES number.	Noted
(Page 3 of 5) 	COMPANY Comment is noted and document has been updated accordingly. However COMPANY to note that we are working on COMPANY Live SPI database where the existing PSR is following for the project scope. We have used D-SC as DCS serial communication(There is no Analog and Digital for serial communication I/O in the SPI Database)	Noted



COMMENTS RESOLUTION SHEET

ADNOC GAS

NM-30201-50200-H0-113-13-15-25-001 C CRS



HABSHAN & DIVERT EXCESS MP FUEL GAS PROJECT

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COMMENTS RESOLUTION SHEET

ADNOC GAS

12.	(Page 3 of 5) COMPANY Comment is noted and document has been updated accordingly. PAHH will be in DCS through serial communication from ESD system Refer Marked up Comments
13.	(Page 3 of 5) COMPANY Comment is noted and document has been updated accordingly. However COMPANY to note that we are working on COMPANY Live SPI database where the existing PSR is following for the project scope. Partial can't be updated for Instrument profile as it will affect existing Company live SPI database, However it is added in Instrument Discription. Refer Marked Up Comments



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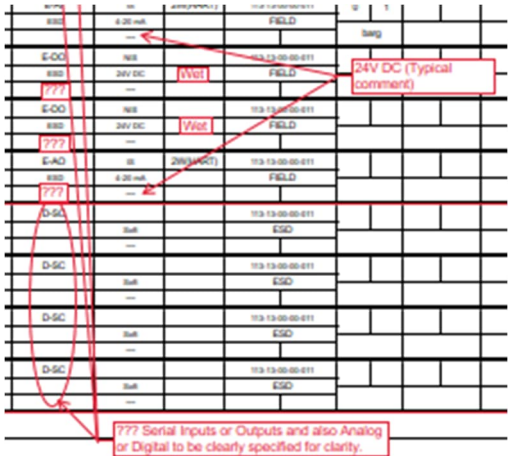
RAD PROJECT No: 5900982

AREJLERS



COMMENTS RESOLUTION SHEET

ADNOC GAS

16.	(Page 3 of 5) 	COMPANY Comment is noted and document has been updated accordingly. However COMPANY to note that we are working on COMPANY Live SPI database where the existing PSR is following for the project scope. ??? please note that this column is for Loop name only as per Approved Company live SPI database template.	Noted
-----	--	---	-------



HABSHAN & DIVERT EXCESS MP FUEL GAS PROJECT
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COMMENTS RESOLUTION SHEET

ADNOC GAS

18.	(Page 3 of 5) 777 Serial Inputs or Outputs and also Analog or Digital to be clearly specified for clarity. Legends and Abbreviations is incomplete. It shall cover specifically all I/O types indicated such as D-AI, D-AO, D-DO, E-AI, E-AO, E-DI, E-DO, Serial I/O's of each type (digital & analog) for each system such as D-SDI, D-SDO, E-SDI, E-SDO etc., specifically which shall replace D-SC indicated. (Typical comment) All HMI Tags shall be indicated as applicable for completeness. (Typical comment) Are these powered outputs? This shall be Potential Free Contacts (PFC). Clarify & correct. (Typical comment) 24V DC	COMPANY Comment is noted and document has been updated accordingly. Legends and Abbreviations are updated, However COMPANY to note that we are working on COMPANY Live SPI database where the existing PSR is following for the project scope Serial digital and analog IO's are not available in the existing Company SPI database.	Noted
COMMENT ON REV.A			
1.	(Page 2) Interface All Holds shall be removed for final approval 777 Clarify, Poor Quality Checks. Correct it as applicable. In line with existing plant philosophy	COMPANY Comment is noted and document has been updated accordingly.	Noted



HABSHAN & DIVERT EXCESS MP FUEL GAS PROJECT
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 RAD PROJECT No: 5900982



COMMENTS RESOLUTION SHEET

ADNOC GAS

2.	(Page 2) shall be carried out in other project(HBCSU_P5323) 3) Serial interface signals shall be configured/mapped on the existing : However, all required Project SOW inputs such as I-O List, Alarm-Trip List, Cause & Effect Diagram, Operating & Control Philosophy, Graphic Inputs, Control Narrative etc., shall be provided by the EPC Contractor/Detail Design Consultant to the Interface Project (HBCSU) for proper implementation of Project SOW meeting all functional & interface requirements. Also all field side work from supply of Field instruments & Control Valve etc., and cabling/wiring the same until DCS Marshalling cabinet shall be under this project SOW by the EPC Contractor/Detail Design Consultant. Also necessary scope in Loop Checking, assistance and participation in FAT, SAT/ISAT etc., of DCS as required for completion of this Project SOW shall be by EPC Contractor/Detail Design Consultant of this project. All required interface with HBCSU Project shall be EPC Contractor/Detail Design Consultant scope/responsibility of this project with Company Support. (Add this also in this note and also confirm the same in CRS)	COMPANY Comment is noted and document has been updated according to Tender Bulletin no.1 dated 27-Feb-24 (TB01).	Noted
3.	(Page 2) 01-00-30-001 Rev.3 SH11 TO 11 - Legend P&ID sheet This is yet to be issued with applicable Project SOW modification. Other P&IDs with Legend reference will not be approved without issue of this Legend P&IDs Include Instrument Index, Instrument Data sheets, Instrument Cable Block Diagram, Instrument & JB Location layout, Instrument Loop Drawing also in Reference Documents & Drawings	i) COMPANY Comment is noted and document has been updated accordingly. ii) As there is no modification or change in legend and considering as small job, we follow existing legend sheet. Project P&ID Legend sheet is not envisaged.	Noted
4.	(Page 2) data.However same shall be verified during detailed Vendor engi FEED data.However same shall be verified during detailed Vend All Holds shall be removed for final approval	COMPANY Comment is noted.	Noted
5.	(Page 2) I/O Summary indicated is just noted for information only. It shall be total responsibility of EPC Contractor to check & confirm the completeness & correctness of the I-O summary indicated covering all interface & functional requirements of project scope as per all approved Project inputs. (Typical comment)	COMPANY Comment is noted.	Noted



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COMMENTS RESOLUTION SHEET

ADNOC GAS

6.	(Page 3) Instrument I/O List Format (Fields of information) shall be in line with latest Company approved existing format of recent ADNOC Gas Project. Refer attached supplementary file for reference format to be followed (Typical comment)	COMPANY Comment is noted and Instrumentv I/O list document has been updated according to company furnished I/O list template.	Noted
7.	(Page 3) Instrument Type No. 	COMPANY Comment is noted and document has been updated accordingly.	Noted
8.	(Page 3) Include Loop No. also in the I-O List.	COMPANY Comment is noted and loop number column has been updated accordingly.	Noted



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COMMENTS RESOLUTION SHEET

ADNOC GAS

9.	(Page 3) <table><tr><td></td><td></td><td>from U/S of 90-V-101 (Typical comment)</td></tr><tr><td></td><td>Service Description</td><td>P & ID No.</td></tr><tr><td></td><td>MP Fuel Gas Pressure</td><td>30201-50200-H0-113-13-00-00</td></tr><tr><td></td><td>MP Fuel Gas Pressure</td><td>30201-50200-H0-113-13-00-00</td></tr><tr><td></td><td>MP Fuel Gas Pressure</td><td>30201-50200-H0-113-13-00-00</td></tr><tr><td>MP</td><td>MP Fuel Gas Pressure</td><td>30201-50200-H0-113-13-00-00</td></tr><tr><td>1</td><td>MP Fuel Gas Pressure</td><td>30201-50200-H0-113-13-00-00</td></tr></table>			from U/S of 90-V-101 (Typical comment)		Service Description	P & ID No.		MP Fuel Gas Pressure	30201-50200-H0-113-13-00-00		MP Fuel Gas Pressure	30201-50200-H0-113-13-00-00		MP Fuel Gas Pressure	30201-50200-H0-113-13-00-00	MP	MP Fuel Gas Pressure	30201-50200-H0-113-13-00-00	1	MP Fuel Gas Pressure	30201-50200-H0-113-13-00-00	COMPANY Comment is noted and service description has been updated accordingly.	Noted
		from U/S of 90-V-101 (Typical comment)																						
	Service Description	P & ID No.																						
	MP Fuel Gas Pressure	30201-50200-H0-113-13-00-00																						
	MP Fuel Gas Pressure	30201-50200-H0-113-13-00-00																						
	MP Fuel Gas Pressure	30201-50200-H0-113-13-00-00																						
MP	MP Fuel Gas Pressure	30201-50200-H0-113-13-00-00																						
1	MP Fuel Gas Pressure	30201-50200-H0-113-13-00-00																						
10.	(Page 3) <table><tr><td></td><td>Hand (Typical comment)</td></tr><tr><td>Temperature Indication</td><td>MP Fuel G</td></tr><tr><td>Field Switch-Open</td><td>113-XV-9200 valve open I FI</td></tr><tr><td>Field Switch-Close</td><td>113-XV-9200 valve close</td></tr></table>		Hand (Typical comment)	Temperature Indication	MP Fuel G	Field Switch-Open	113-XV-9200 valve open I FI	Field Switch-Close	113-XV-9200 valve close	COMPANY Comment is noted and hand switch has been updated accordingly.	Noted													
	Hand (Typical comment)																							
Temperature Indication	MP Fuel G																							
Field Switch-Open	113-XV-9200 valve open I FI																							
Field Switch-Close	113-XV-9200 valve close																							



HABSHAN & DIVERT EXCESS MP FUEL GAS PROJECT

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COMMENTS RESOLUTION SHEET

ADNOC GAS

11.	(Page 3) COMPANY Comment is noted and all red marked comment has been updated accordingly.	Noted
12.	(Page 3) COMPANY Comment is noted and document has been updated accordingly.	Noted
13.	(Page3) COMPANY Comment is noted and document has been updated accordingly.	Noted



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COMMENTS RESOLUTION SHEET

ADNOC GAS

14.	(Page 3) <table><tr><td>DO</td><td>Potential Free Contact</td><td>-</td><td>DCS</td><td>ESD</td><td>-</td></tr><tr><td>AI</td><td>4-20 mA</td><td>IS</td><td>Field</td><td>ESD</td><td>2W (HART)</td></tr><tr><td>DO</td><td>Powered Contact</td><td>NIS</td><td>ESD</td><td>Field</td><td>-</td></tr><tr><td>DO</td><td>Powered Contact</td><td>NIS</td><td>ESD</td><td>Field</td><td>-</td></tr><tr><td>AO</td><td>4-20 mA</td><td>IS</td><td>ESD</td><td>Field</td><td>2W (HART)</td></tr></table>	DO	Potential Free Contact	-	DCS	ESD	-	AI	4-20 mA	IS	Field	ESD	2W (HART)	DO	Powered Contact	NIS	ESD	Field	-	DO	Powered Contact	NIS	ESD	Field	-	AO	4-20 mA	IS	ESD	Field	2W (HART)	COMPANY Comment is noted and I/O list updated accordingly.	Noted
DO	Potential Free Contact	-	DCS	ESD	-																												
AI	4-20 mA	IS	Field	ESD	2W (HART)																												
DO	Powered Contact	NIS	ESD	Field	-																												
DO	Powered Contact	NIS	ESD	Field	-																												
AO	4-20 mA	IS	ESD	Field	2W (HART)																												
15.	(Page 3) 16 Indicate the HH set point here also Temp. & Press. Compensated Flow Calculation Block Adder Block (Compensated Flow Adder block for F-1800 & F-1051A) Temp. & Press. Compensated Flow Calculation Block 1	COMPANY Comment is noted and I/O list updated accordingly.	Noted																														
16.	(Page3) Indicate the HH set point here also 15 16	COMPANY Comment is noted and I/O list updated accordingly.	Noted																														



HABSHAN & DIVERT EXCESS MP FUEL GAS PROJECT

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RAD PROJECT No: 5900982

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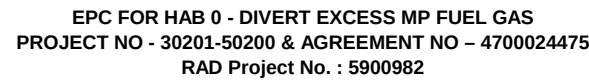


COMMENTS RESOLUTION SHEET

ADNOC GAS

17.	(Page3) All Holds shall be removed for final approval typical comment) <table><tr><th rowspan="2">Field Inst. / transmitter type</th><th rowspan="2">Range (HOLD-1)</th><th rowspan="2">Engg. Unit</th><th colspan="3">Alarm Set Point (HOLD)</th></tr><tr><th>L</th><th>LL</th><th>H</th></tr><tr><td></td><td></td><td></td><td></td><td></td><td></td></tr></table>	Field Inst. / transmitter type	Range (HOLD-1)	Engg. Unit	Alarm Set Point (HOLD)			L	LL	H							COMPANY Comment is noted.	Noted																													
Field Inst. / transmitter type	Range (HOLD-1)				Engg. Unit	Alarm Set Point (HOLD)																																									
		L	LL	H																																											
18.	(Page 3) DOC. NO.: NM-30201-50200-H0-113-13-15-25-001 TRACTOR DOC. NO.: 5900892-4C-LST-0001 DC Loop Powered (Typical comment) <table><tr><th rowspan="2">Signal Type</th><th rowspan="2">Signal</th><th colspan="2">Signal</th><th rowspan="2">Field Inst. / Transmitter type</th><th rowspan="2">Range (HOLD)</th></tr><tr><th>From</th><th>To</th></tr><tr><td>4-20 mA</td><td>IS</td><td>Field</td><td>DCS</td><td>2W (HART)</td><td>0 - 40</td></tr><tr><td>Soft</td><td>-</td><td>-</td><td>-</td><td>-</td><td>-</td></tr><tr><td>4-20 mA</td><td>IS</td><td>DCS</td><td>Field</td><td>2W (HART)</td><td>-</td></tr><tr><td>4-20 mA</td><td>IS</td><td>Field</td><td>DCS</td><td>2W (HART)</td><td>0-100</td></tr><tr><td>Soft</td><td>-</td><td>-</td><td>-</td><td>-</td><td>-</td></tr><tr><td>4-20 mA</td><td>IS</td><td>Field</td><td>DCS</td><td>2W (HART)</td><td>0 - 20</td></tr></table>	Signal Type	Signal	Signal		Field Inst. / Transmitter type	Range (HOLD)	From	To	4-20 mA	IS	Field	DCS	2W (HART)	0 - 40	Soft	-	-	-	-	-	4-20 mA	IS	DCS	Field	2W (HART)	-	4-20 mA	IS	Field	DCS	2W (HART)	0-100	Soft	-	-	-	-	-	4-20 mA	IS	Field	DCS	2W (HART)	0 - 20	As per COMPANY furnished SPI template followed and we used 4-20 mA only otherwise importing time error will come in SPI.	Noted
Signal Type	Signal			Signal				Field Inst. / Transmitter type	Range (HOLD)																																						
		From	To																																												
4-20 mA	IS	Field	DCS	2W (HART)	0 - 40																																										
Soft	-	-	-	-	-																																										
4-20 mA	IS	DCS	Field	2W (HART)	-																																										
4-20 mA	IS	Field	DCS	2W (HART)	0-100																																										
Soft	-	-	-	-	-																																										
4-20 mA	IS	Field	DCS	2W (HART)	0 - 20																																										
19.	(Page 3) Include "Line / Equipment No." Field also EPC FOR HAB 0 - DIVERT EXCESS MP FUEL G PROJECT NO - 30201-50200 & AGREEMENT NO - 47/ RAD Project No.: 5900982 INSTRUMENT I/O LIST COMPANY DOC. NO.: NM-30201-50200-H0-113-13-15- CONTRACTOR DOC. NO.: 5900892-4C-LST-0001 <table><tr><th>I/O Type</th><th>Signal Type</th><th>Signal</th></tr><tr><td></td><td></td><td></td></tr></table>	I/O Type	Signal Type	Signal				As per COMPANY furnished template of Instrument I/O list there is no Line / Equipment no field.	Noted																																						
I/O Type	Signal Type	Signal																																													

	EPC FOR HAB 0 - DIVERT EXCESS MP FUEL GAS PROJECT NO - 30201-50200 & AGREEMENT NO – 4700024475 RAD Project No. : 5900982				 	
	INSTRUMENT I/O LIST					
ADNOC GAS	COMPANY DOC. NO.: NM-30201-50200-H0-113-13-15-25-001				REV	0
	CONTRACTOR DOC. NO.: 5900892-IC-LST-0001				DATE	12-12-2025
EPC FOR HAB 0 - DIVERT EXCESS MP FUEL GAS INSTRUMENT I/O LIST						
						
0	12/12/2025	ISSUED FOR CONSTRUCTION		UVS	JSB	MMM
D	14-10-2025	RE-ISSUED FOR APPROVAL (IFA)		UVS	SKD	MMM
C	22-04-2025	RE-ISSUED FOR APPROVAL (IFA)		RKB	SKD	MMM
B	28-02-2025	ISSUED FOR APPROVAL (IFA)		MMS	SKD	MMM
A	06-11-2024	ISSUED FOR REVIEW (IFR)		MMS	SKD	MMM
REV.	DATE	REVISION DESCRIPTION		PRPD	CHKD	APPD
Doc. Class: 2		Company Doc. No.: NM-30201-50200-H0-113-13-15-25-001		Contractor Doc. No.: 5900892-IC-LST-0001		Rev.0



COMPANY DOC. NO.: NM-30201-50200-H0-113-13-15-25-001

0

12-12-2025

SHT NO.	DESCRIPTION
1	COVER SHEET
2	INDEX, GENERAL NOTES, ABBREVIATION & REFERENCE DOCUMENT
3,4	IO LIST

- 1) All ESD signals shall be terminated in existing junction box as per JB termination DW-01-15-35-039 for IES-15, 15-ES-02 ESD system interfacing
- 2) All DCS signals shall be terminated in IES-15 NEW DCS Marshalling Cabinet . Existing DCS (ABB) is getting replaced with new Yokogawa Centum VP DCS as part of on-going project (P5323, HBCSU). Hence, there is no requirement of engaging the existing OEM -DCS (ABB). DCS (M/s Yokogawa) modifications (addition of IO modules, configuration of graphics, logics pertaining to this project) shall be carried out in other interface project (HBCSU, P5323). EPC Contractor to accommodate this project requirements in new DCS systems.

EPC Contractor shall include cable termination from field up to the marshalling cabinets in IES, loop checks, provision of necessary assistance to functional logics/graphics (FAT/IFAT/SAT) checks related in their scope.
- 3) Serial interface signals shall be configured/mapped on the existing serial interface between ESD and DCS in line with existing plant philosophy.
- 4) [Alarm Priorities and Rationalization shall be done for the Project SOW in line with existing Plant Philosophy](#)

Document No.	Title
BOD-30201-50200-H0-113-13-15-00-001	INSTRUMENTATION DESIGN BASIS
30201-50200-H0-90-00-0301-A1	P&ID AREA 90 FUEL GAS & FLARE SYSTEM-CONSTRUCTION
30201-50200-H0-113-13-00-00-011-A1	P&ID DIVERT EXCESS MP FUEL GAS LINE CONSTRUCTION
30201-50200-H0-113-00-30-003-002-A1	P&ID MP GAS MANIFOLD UNIT 113 (NORTH AREA)- CONSTRUCTION
30201-50200-H0-13-00-213-A1	P&ID FLARE HEADER EARLY THAMMAMA 'B' (HABSHAN)-CONSTRUCTION
30201-50200-H0-13-00-213D-A1	P&ID FLARE HEADER EARLY THAMMAMA 'B' (HABSHAN)-DEMOLITION
30201-50200-H0-90-00-0301D-A1	P&ID AREA 90 FUEL GAS & FLARE SYSTEM-DEMOLITION
30201-50200-H0-113-00-30-003-002D-A1	P&ID MP GAS MANIFOLD UNIT 113 (NORTH AREA)-DEMOLITION
01-00-30-001 Rev.3 SHT1 TO 11	EXISTING LEGEND P&ID SHEET
SPP-30201-50200-H0-113-13-15-00-002	DATA SHEET FOR CONTROL VALVE
SPP-30201-50200-H0-113-13-15-00-003	DATA SHEET FOR SHUT DOWN VALVE
SPP-30201-50200-H0-113-13-15-70-001	DATA SHEET FOR PRESSURE TRANSMITTER
SPP-30201-50200-H0-113-13-15-70-002	DATA SHEET FOR FLOW TRANSMITTER
SPP-30201-50200-H0-113-13-15-70-003	DATA SHEET FOR ORIFICE PLATES
SPP-30201-50200-H0-113-13-15-70-005	DATASHEET FOR DIFFERENTIAL PRESSURE TRANSMITTER
SPP-30201-50200-H0-113-13-15-70-007	DATASHEET FOR TEMPERATURE TRANSMITTER, TEMPERATURE ELEMENT WITH THERMOWELL
30201-50200-H0-113-13-15-00-001-A3	INSTRUMENT CABLE BLOCK DIAGRAM
DWG-30201-50200-H0-113-13-15-80-001	INSTRUMENT LOOP DIAGRAMS
30201-50200-H0-113-13-15-41-002-A1	INSTRUMENT AND JB LOCATION LAYOUTS
30201-50200-H0-113-13-15-39-002-A3	INSTRUMENT INSTALLATION DRAWINGS
30201-50200-H0-113-13-15-38-001-A3	INSTRUMENT HOOK UP DRAWINGS
DOD-30201-50200-H0-113-13-15-00-011	INSTRUMENT INDEX
DW-01-15-35-039	JB Termination Drawing
SDOC-5225-01-1510-0010-0036	I/O system Map-Modus interchange Table IES 15 Unit 113(N)
DOD-30201-50200-H0-113-13-00-00-006	Alarm and Trip List

I/O SUMMARY LIST:

	DCS Signals	ESD Signals
I/O Type	Qty.	Qty.
AI (IS)	5	3
AO (IS)	1	-
DI (IS)	-	4
DI(NIS)	-	4
DO(NIS)	4	2

Rev. 0



HABSHAN & DIVERT EXCESS MP FUEL GAS PROJECT

INSTRUMENT I/O LIST

Document No.	: 30201-50200-H0-113-13-15-25-001	Revision	: 0
Plant	: Habshan4_HGCE	Revision Description	: INSTRUMENT I/O LIST
Area	: COMPRESSION	Page	: 1 of 1
Unit	: 113	Date	: 12/11/2025
IES Number	: DCS		

S. No.	Tag Number	Instrument Type	From	I/O Type	IS / NIS	Wire Type	P&ID No		Sys. Range		Alarm / Trip Settings				Alarm	Marsh Cab No.	JB Number	Pri Cable No	Remarks	Rev
	Loop Number	Service Description	To	System	Signal Type	Wet / Dry	Device Location		Min.	Max.	H	HH	L	LL	Priority	Sys Cab. No.	Intercon. Dwg. No.	Cable Size		
		HMI Description	Status	HMI Tag	Voltage LVL	NO / NC	MOS	OOS	Unit	State Text 0	State Text 1	Voting No	IO Group No.	Loop Dwg. No.	Pr./Tr./Core					
1	113-FT -1851	FLOW DP TRANSMITTER	FIELD	D-AI	IS	2W(HART)	113-13-00-00-011		0	2500	---	---	---	---	NA	15-DT-01	113-A-16101	113A16101	New Junction Box and Multipair Cable to be used	0
	113-F-1851	MP Fuel Gas from U/S of 90-V-101	DCS	DCS	4-20 mA	---	FIELD		mmH2O				---		15-DS-01	DW-01-15-35-039	10PX1.00 sqmm			
		MP Fuel Gas from U/S of 90-V-101	N-NEW	113-FI -1851	24V DC	---									113-13-15-80-001	PR5				
2	113-PT -3191	Pressure Transmitter	FIELD	D-AI	IS	2W(HART)	113-13-00-00-011		0	40	---	---	---	---	NA	15-DT-01	113-A-16101	113A16101	New Junction Box and Multipair Cable to be used	0
	113-P-3191	MP Fuel Gas from U/S of 90-V-101	DCS	DCS	4-20 mA	---	FIELD		BG-gg				---		15-DS-01	DW-01-15-35-039	10PX1.00 sqmm			
		MP Fuel Gas from U/S of 90-V-101	N-NEW	113-PI -3191	24V DC	---									113-13-15-80-001	PR1				
3	113-PY -3191	VALVE POSITIONER	DCS	D-AO	IS	2W(HART)	113-13-00-00-011				---	---	---	---	NA	15-DT-01	113-A-16101	113A16101	New Junction Box and Multipair Cable to be used	0
	113-P-3191	MP Fuel Gas from U/S of 90-V-101	FIELD	DCS	4-20 mA	---	FIELD						---		15-DS-01	DW-01-15-35-039	10PX1.00 sqmm			
		MP Fuel Gas from U/S of 90-V-101	N-NEW	113-PY -3191	24V DC	---									113-13-15-80-001	PR2				
4	113-PZT -3191	Position Transmitter	FIELD	D-AI	IS	2W(HART)	113-13-00-00-011		0	100	---	---	---	---	NA	15-DT-01	113-A-16101	113A16101	New Junction Box and Multipair Cable to be used	0
	113-P-3191	MP Fuel Gas from U/S of 90-V-101	DCS	DCS	4-20 mA	---	FIELD		%				---		15-DS-01	DW-01-15-35-039	10PX1.00 sqmm			
		MP Fuel Gas from U/S of 90-V-101	N-NEW	113-PZI -3191	24V DC	---									113-13-15-80-001	PR3				
5	113-PT -3192	Pressure Transmitter	FIELD	D-AI	IS	2W(HART)	113-13-00-00-011		0	25	15.4	---	---	---	High	15-DT-01	113-A-16101	113A16101	New Junction Box and Multipair Cable to be used	0
	113-P-3192	MP Fuel Gas to MP Gas Manifold	DCS	DCS	4-20 mA	---	FIELD		barg				---		15-DS-01	Dw-01-15-35-039	10PX1.00 sqmm			
		MP Fuel Gas to MP Gas Manifold	N-NEW	113-PI -3192	24V DC	---									113-13-15-80-001	PR4				
6	113-TT -5101	Temperature Transmitter	FIELD	D-AI	IS	2W(HART)	113-13-00-00-011		-4	85	---	---	---	---	TBC	15-DT-01	113-A-16101	113A16101	New Junction Box and Multipair Cable to be used	0
	113-T-5101	MP Fuel Gas Temperature	DCS	DCS	4-20 mA	---	FIELD		°C				---		15-DS-01	DW-01-15-35-039	10PX1.00 sqmm			
		MP Fuel Gas Temperature	N-NEW	113-TI -5101	24V DC	---	-									113-13-15-80-001	PR6			
7	113-XHSC -9501B	DCS HAND SWITCH CLOSE	DCS	D-DO	NIS	---	113-13-00-00-011				---	---	---	---			---		Hardwire signal B/W DCS and ESD system	0
	113-X-9501B	113-XV-9501 Valve Close Command From DCS	ESD	DCS	POTENTIAL FREE	---	DCS						---		15-DS-01	---				
		113-XV-9501 Valve Close Command From DCS	N-NEW	113-XHSC -9501B	---	---									113-13-15-80-001					
8	113-XHSO -9501B	DCS HAND SWITCH OPEN	DCS	D-DO	NIS	---	113-13-00-00-011				---	---	---	---			---		Hardwire signal B/W DCS and ESD system	0
	113-X-9501B	113-XV-9501 Valve Open Command From DCS	ESD	DCS	POTENTIAL FREE	---	DCS						---		15-DS-01	---				
		113-XV-9501 Valve Open Command From DCS	N-NEW	113-XHSO-9501B	---	---									113-13-15-80-001					
9	113-XHSC -9502B	DCS HAND SWITCH CLOSE	DCS	D-DO	NIS	---	113-13-00-00-011				---	---	---	---			---		Hardwire signal B/W DCS and ESD system	0
	113-X-9502B	113-XV-9502 Valve close Command From DCS	ESD	DCS	POTENTIAL FREE	---	DCS						---		15-DS-01	---				
		113-XV-9502 Valve close Command From DCS	N-NEW	113-XHSC -9502B	---	---									113-13-15-80-001					
10	113-XHSO -9502B	DCS HAND SWITCH OPEN	DCS	D-DO	NIS	---	113-13-00-00-011				---	---	---	---			---		Hardwire signal B/W DCS and ESD system	0
	113-X-9502B	113-XV-9502 Valve Open Command From DCS	ESD	DCS	POTENTIAL FREE	---	DCS						---		15-DS-01	---				
		113-XV-9502 Valve Open Command From DCS	N-NEW	113-XHS0-9502B	---	---									113-13-15-80-001					



HABSHAN & DIVERT EXCESS MP FUEL GAS PROJECT

INSTRUMENT I/O LIST

Document No.	: 30201-50200-H0-113-13-15-25-001	Revision	: 0
Plant	: Habshan4_HGCE	Revision Description	: INSTRUMENT I/O LIST
Area	: COMPRESSION	Page	: 1 of 1
Unit	: 113	Date	: 12/11/2025
IES Number	: ESD		

S. No.	Tag Number	Instrument Type	From	I/O Type	IS / NIS	Wire Type	P&ID No		Sys. Range		Alarm / Trip Settings				Alarm Priority	Marsh Cab No.	JB Number	Pri Cable No	Remarks	Rev
	Loop Number	Service Description	To	System	Signal Type	Wet / Dry	Device Location		Min.	Max.	H	HH	L	LL		Sys Cab. No.	Intercon. Dwg. No.	Cable Size		
		HMI Description	Status	HMI Tag	Voltage LVL	NO / NC	MOS	OOS	Unit	State Text 0	State Text 1	Voting No	IO Group No.	Loop Dwg. No.	Pr./Tr./Core					
1	113-PT -3193A	Pressure Transmitter	FIELD	E-AI	IS	2W(HART)	113-13-00-00-011		0	20	---	16	---	---	High	15-ET-02B	113-A-15018	113A15018	Existing Multipair Cable and Junction Box to be Utilised	0
	113-P-3193	MP Fuel Gas from U/S of 90-V-101	ESD	ESD	4-20 mA	---	FIELD		barg			1 OO 2	15-ES-02	DW-01-15-35-039	10PX1.00 sqmm					
		MP Fuel Gas from U/S of 90-V-101	N-NEW	113-PI -3193A	24V DC	---							113-13-15-80-001	PR6						
2	113-PT -3193B	Pressure Transmitter	FIELD	E-AI	IS	2W(HART)	113-13-00-00-011		0	20	---	16	---	---	High	15-ET-02B	113-A-15015	113A15015	Existing Multipair Cable and Junction Box to be utilised	0
	113-P-3193	MP Fuel Gas to MP Gas Manifold	ESD	ESD	4-20 mA	---	FIELD		barg			1 OO 2	15-ES-02	DW-01-15-35-039	20PX1.00 sqmm					
		MP Fuel Gas to MP Gas Manifold	N-NEW	113-PI -3193B	24V DC	---							113-13-15-80-001	PR14						
3	113-PDT -3194	Pressure Differential Transmitter	FIELD	E-AI	IS	2W(HART)	113-13-00-00-011		0	2	---	1	---	---	High	15-ET-02B	113-A-15015	113A15015	Existing Multipair Cable and Junction Box to be Utilised	0
	113-P-3194	MP Fuel Gas Differential Pressure	ESD	ESD	4-20 mA	---	FIELD		barg			---	15-ES-02	DW-01-15-35-039	20PX1.00 sqmm					
		MP Fuel Gas Differential Pressure	N-NEW	113-PDI -3194	24V DC	---							113-13-15-80-001	PR13						
4	113-XHSC -9501A	FIELD HAND SWITCH CLOSE	FIELD	E-DI	NIS	---	113-13-00-00-011				---	---	---	---		15-ET-03B	113-D-15005	113D15005	Existing JB and Multipair cable to be utilised	0
	113-X-9501	113-XV-9501 Valve Close HS from Field	ESD	ESD	POTENTIAL FREE	---	FIELD					---	15-ES-02	DW-01-15-35-039	30PX1.00 sqmm					
		113-XV-9501 Valve Close HS from Field	N-NEW	113-XHSC -9501A	---	---							113-13-15-80-001	PR14						
5	113-XHSO -9501A	FIELD HAND SWITCH OPEN	FIELD	E-DI	NIS	---	113-13-00-00-011				---	---	---	---		15-ET-03B	113-D-15005	113D15005	Existing JB and Multipair cable to be utilised	0
	113-X-9501	113-XV-9501 Valve open HS From Field	ESD	ESD	POTENTIAL FREE	---	FIELD					---	15-ES-02	DW-01-15-35-039	30PX1.00 sqmm					
		113-XV-9501 Valve open HS From Field	N-NEW	113-XHSO -9501A	---	---							113-13-15-80-001	PR13						
6	113-XY -9501	SOLENOID VALVE.	ESD	E-DO	NIS	---	113-13-00-00-011				---	---	---	---		15-ET-02A	113-C-15008	113C15008	Existing Multipair Cable and Junction Box to be Utilised	0
	113-X-9501	3-way ESD trip Solenoid valve	FIELD	ESD	24V DC	Wet	FIELD					---	15-ES-02	DW-01-15-35-039	10PX2.50 sqmm					
		3-way ESD trip Solenoid valve	N-NEW	113-XY -9501	---	---							113-13-15-80-001	PR5						
7	113-XZSC -9501	POSITION CLOSED PROXIMITY SWITCH	FIELD	E-DI	IS	---	113-13-00-00-011				---	---	---	---		15-ET-03B	113-D-15006	113D15006	Existing JB and Multipair cable to be utilised	0
	113-X-9501	113-XV-9501 Valve Closed Position limit switch	ESD	ESD	POTENTIAL FREE	---	FIELD					---	15-ES-02	DW-01-15-35-039	30PX1.00 sqmm					
		113-XV-9501 Valve Closed Position limit switch	N-NEW	113-XZSC -9501	---	---							113-13-15-80-001	PR18						
8	113-XZSO -9501	POSITION OPEN PROXIMITY SWITCH	FIELD	E-DI	IS	---	113-13-00-00-011				---	---	---	---		15-ET-03B	113-D-15006	113D15006	Existing JB and Multipair cable to be utilised	0
	113-X-9501	113-XV-9501 Valve Open Position limit switch	ESD	ESD	POTENTIAL FREE	---	FIELD					---	15-ES-02	DW-01-15-35-039	30PX1.00 sqmm					
		113-XV-9501 Valve Open Position limit switch	N-NEW	113-XZSO -9501	---	---							113-13-15-80-001	PR17						
9	113-XHSC -9502A	FIELD HAND SWITCH CLOSE	FIELD	E-DI	NIS	---	113-13-00-00-011				---	---	---	---		15-ET-03B	113-D-15016	113D15016	Existing JB and Multipair cable to be utilised	0
	113-X-9502	113-XV-9502 Valve Close HS from Field	ESD	ESD	POTENTIAL FREE	---	FIELD					---	15-ES-02	DW-01-15-35-039	20PX1.00 sqmm					
		113-XV-9502 Valve Close HS from Field	N-NEW	113-XHSC -9502A	---	---							113-13-15-80-001	PR10						
10	113-XHSO -9502A	FIELD HAND SWITCH OPEN	FIELD	E-DI	NIS	---	113-13-00-00-011				---	---	---	---		15-ET-03B	113-D-15016	113D15016	Existing JB and Multipair cable to be utilised	0
	113-X-9502	113-XV-9502 Valve open HS from Field	ESD	ESD	POTENTIAL FREE	---	FIELD					---	15-ES-02	DW-01-15-35-039	20PX1.00 sqmm					
		113-XV-9502 Valve open HS from Field	N-NEW	113-XHSO -9502A	---	---							113-13-15-80-001	PR9						
11	113-XY -9502	SOLENOID VALVE.	ESD	E-DO	NIS	---	113-13-00-00-011				---	---	---	---		15-ET-02A	113-C-15008	113C15008	Existing JB and Multipair cable to be utilised	0
	113-X-9502	3-way ESD Trip Solenoid valve	FIELD	ESD	24V DC	Wet	FIELD					---	15-ES-02	DW-01-15-35-039	10PX2.50 sqmm					
		3-way ESD Trip Solenoid valve	N-NEW	113-XY -9502	---	---							113-13-15-80-001	PR7						
12	113-XZSC -9502	POSITION CLOSED PROXIMITY SWITCH	FIELD	E-DI	IS	---	113-13-00-00-011				---	---	---	---		15-ET-03B	113-D-15006	113D15006	Existing JB and Multipair cable to be utilised	0
	113-X-9502	113-XV- 9502 Valve Close Limit Switch	ESD	ESD	POTENTIAL FREE	---	FIELD					---	15-ES-02	DW-01-15-35-039	30PX1.00 sqmm					
		113-XV- 9502 Valve Close Limit Switch	N-NEW	113-XZSC -9502	---	---							113-13-15-80-001	PR20						
13	113-XZSO -9502	POSITION OPEN PROXIMITY SWITCH	FIELD	E-DI	IS	---	113-13-00-00-011				---	---	---	---		15-ET-03B	113-D-15006	113D15006	Existing JB and Multipair cable to be utilised	0
	113-X-9502	113-XV-9502 Valve Open limit Switch	ESD	ESD	POTENTIAL FREE	---	FIELD					---	15-ES-02	DW-01-15-35-039	30PX1.00 sqmm					
		113-XV-9502 Valve Open limit Switch	N-NEW	113-XZSO -9502	---	---							113-13-15-80-001	PR19						